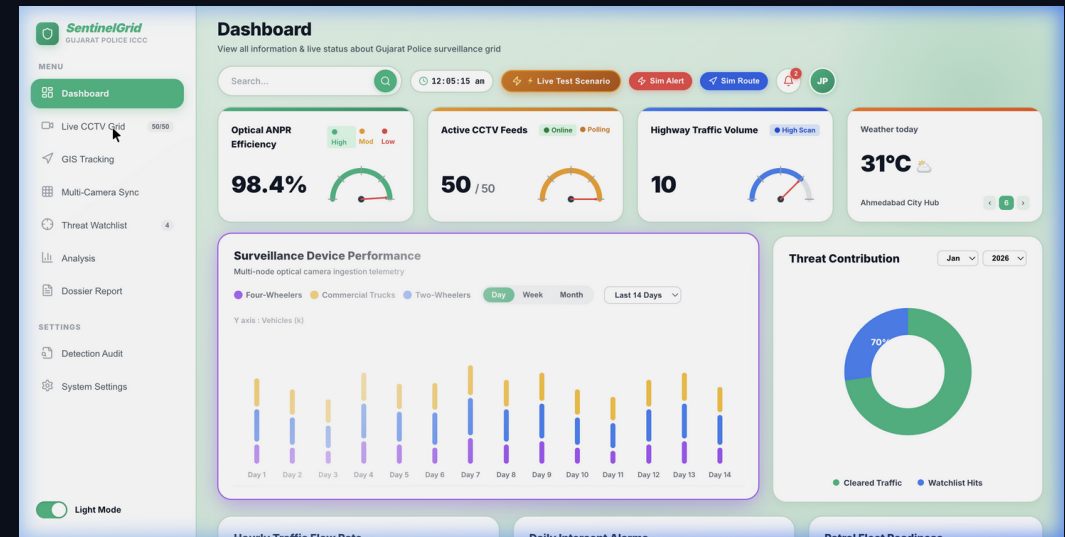


SentinelGrid - Unified Statewide Smart Surveillance

Gujarat Police CCTV Hackathon 2026 | Computer Vision & Vehicle Tracking

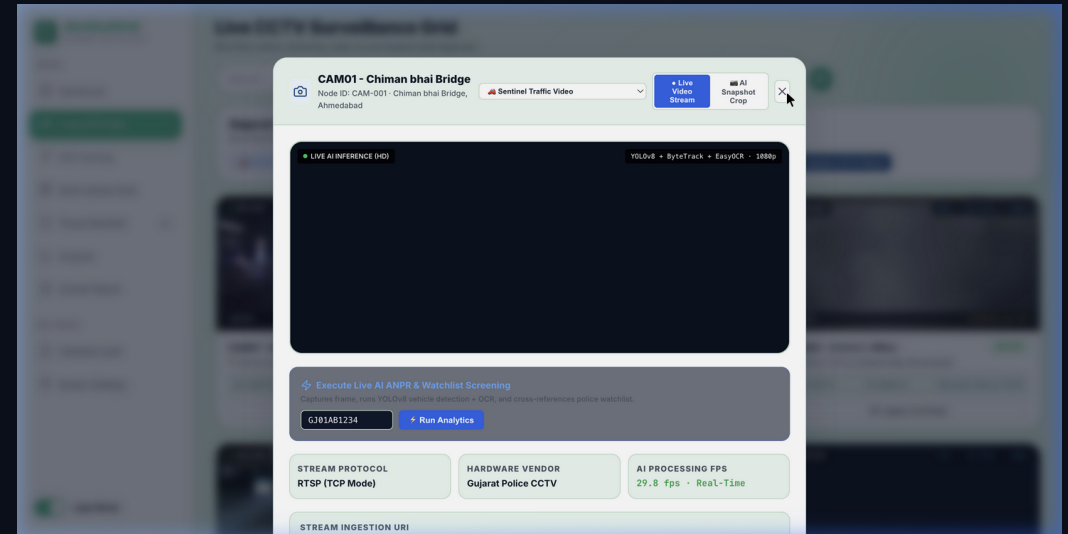
- Track: Computer Vision & Multi-Camera Vehicle Tracking
- Architecture: 3-Tier Hierarchical Edge-Cluster (HEC) Hybrid Architecture
- Scalability: Designed for statewide deployment (approx 80,000 cameras across 33 districts)
- Compliance: Full compliance with Section 65B Indian Evidence Act / Section 63 BSA 2023
- Live Platform: 50 heterogeneous camera feeds integrated across RTSP, WebRTC, and HLS



Slide 2: Problem Statement & Operational Challenges

Critical bottlenecks in statewide multi-agency surveillance

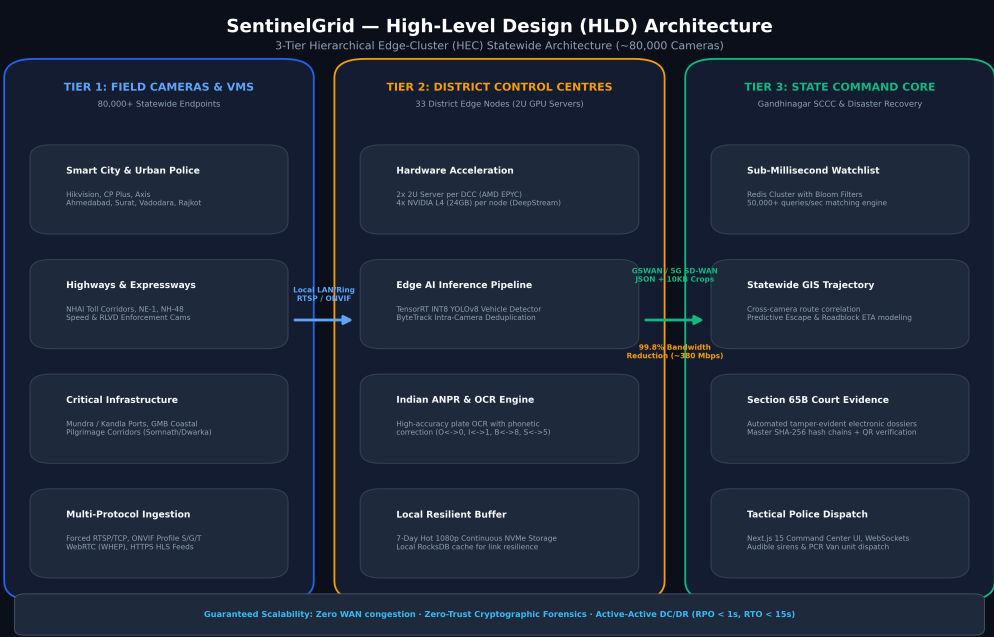
- Siloed VMS Platforms: Incompatible systems (CP Plus, Hikvision, Dahua, Axis, Milestone).
- The 200 Gbps WAN Choke: Centralizing raw video from 80,000 cameras is unviable over GSWAN.
- Jurisdictional Blind Spots: Tracking suspect vehicles across district boundaries is slow and manual.
- Evidentiary Challenges: Uncertified CCTV clips frequently face admissibility hurdles in court.



Slide 3: High-Level Architecture (3-Tier HEC Model)

Edge AI Ingestion + GSWAN Backbone + Gandhinagar SCCC Core

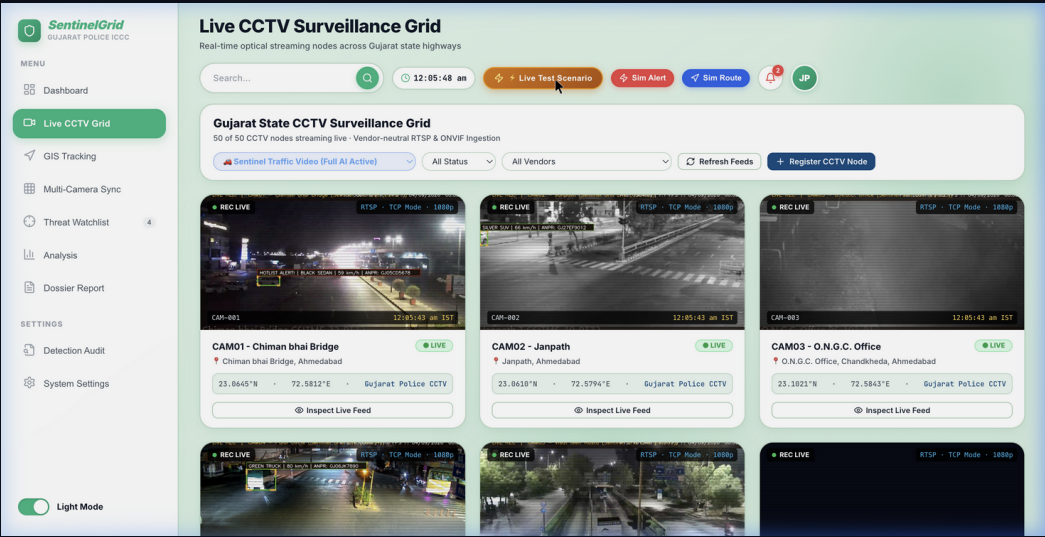
- Tier 1 (Field Edge): Ingestion middleware across RTSP, ONVIF Profile S/G/T, WHEP, and HLS.
- Tier 2 (33 District Control Centres): DeepStream + INT8 TensorRT YOLOv8, ByteTrack & Indian ANPR.
- Tier 3 (State Command Core): Sub-millisecond Redis matching, statewide GIS correlation, and dispatch.
- Bandwidth Savings: 99.8% reduction in WAN bandwidth (approx 380 Mbps vs approx 200 Gbps raw video).



Slide 4: Technical Test Case Compliance (~50 Cameras)

Live sandbox ingestion and multi-feed synchronization

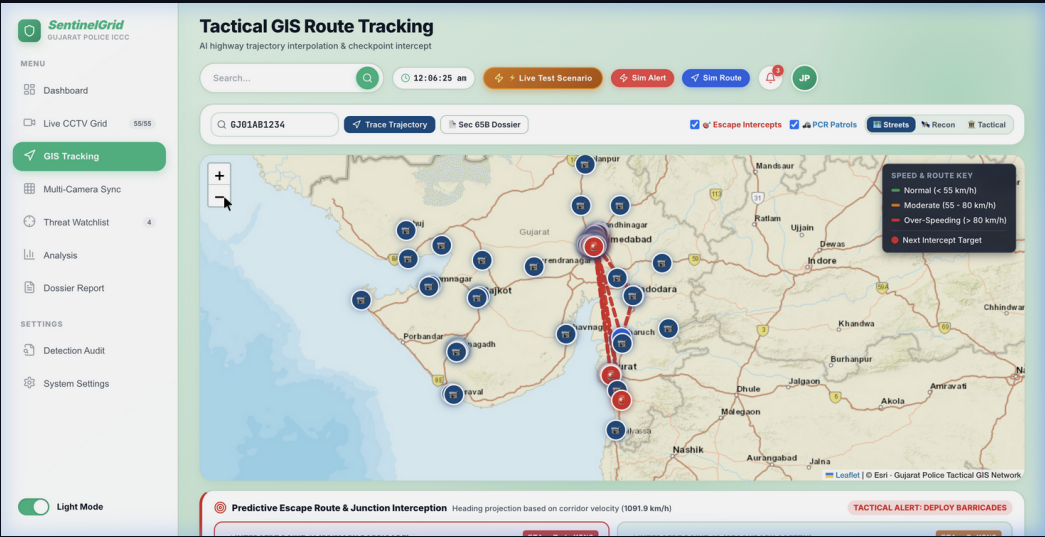
- 50 Heterogeneous Feeds: Ahmedabad SCRB, Surat Smart City, Vadodara NH-48, Rajkot, Mundra Port.
- Monotonic PTS Synchronization: Eliminates clock drift and frame arrival skew across feeds.
- Resilient Networking: Forced RTSP over TCP with exponential backoff auto-reconnect.
- GOP Boundary Handling: Gracefully decodes mid-stream H.264/H.265 join frames without dropping.



Slide 5: Live Vehicle Tracking & Trajectory Reconstruction

Full corridor traversal tracking for designated target GJ01AB1234

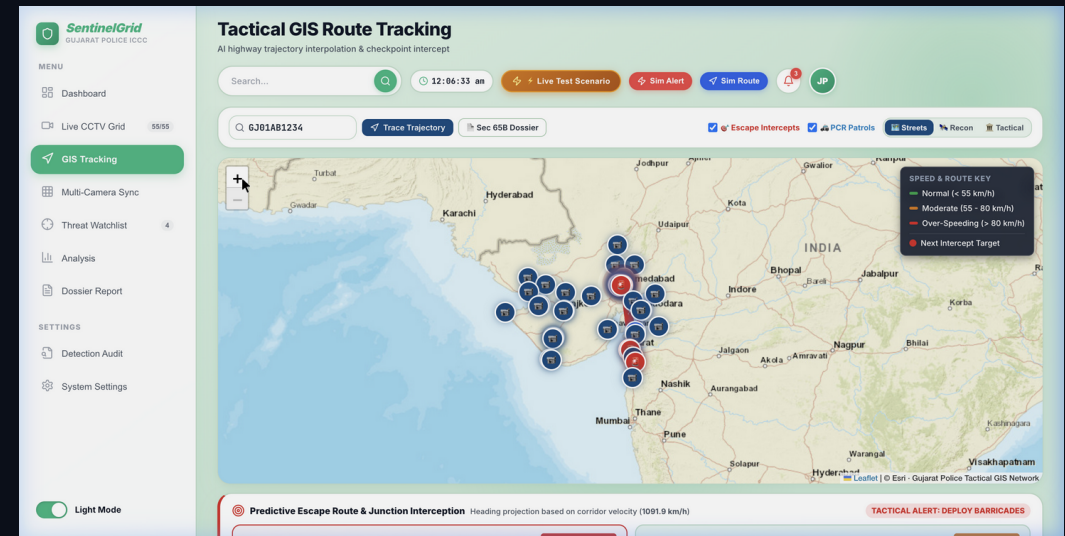
- Target Vehicle: GJ01AB1234 (White Hyundai Creta SUV - Stolen Priority).
- 9 Checkpoints Traced: Chimanbhai Bridge to Paldi to Adalaj to Delight to Mohanpura to Vadodara to Bharuch to Surat to Navsari.
- Telemetry Captured: Body type (SUV), Color (White), Speed (55-84 km/h), Cryptographic PTS hash.
- Sub-Millisecond Screening: In-memory Redis Bloom filter match confirmed in < 1 ms.



Slide 6: AI Predictive Interception & PCR Dispatch

Predictive velocity modeling and tactical roadblock containment

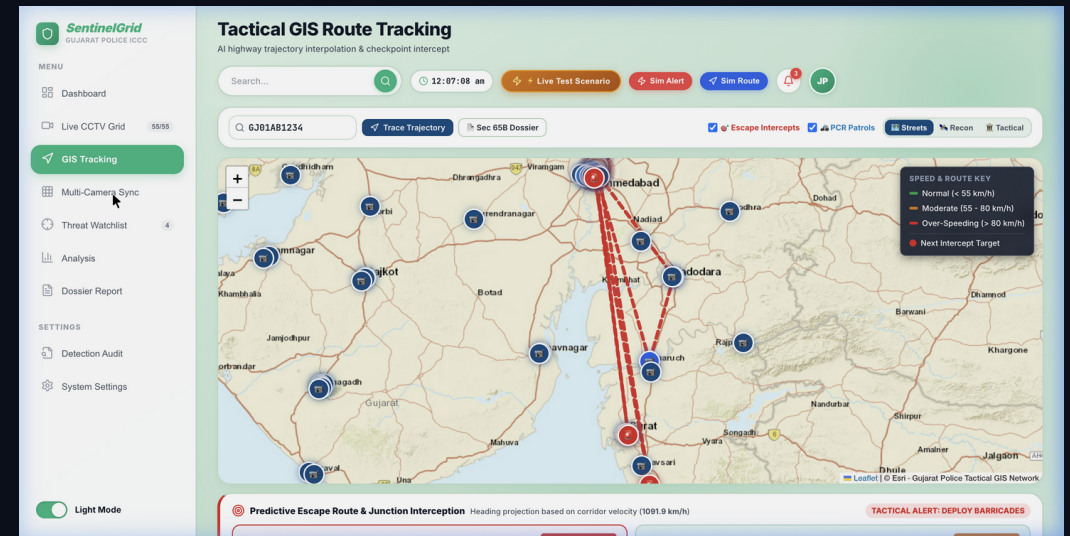
- Predictive Trajectory: Computes heading vector to forecast next downstream road junctions.
- Roadblock ETAs: Predicts exact arrival time at Barricade Alpha and Barricade Bravo.
- Nearest PCR Unit Routing: Automatically indexes and dispatches closest patrol vans (Sagar-22).
- Tactical Siren Dispatch: Browser Web Audio siren broadcasts and voice synthesizer alerts.



Slide 7: Section 65B Indian Evidence Act Admissibility

Cryptographically sealed court-admissible electronic dossiers

- Legal Admissibility: 100% compliant with Section 65B(4) IEA and Section 63 BSA 2023.
- SHA-256 Hash Chain: Master cryptographic hash salted with Camera UUID and PTS timestamp.
- National RTO Integration: Vahan & Sarathi cross-referencing (Chassis, Engine, Owner, Active FIR).
- Instant Verification: Tamper-evident QR code for judicial inspection.



Slide 8: Statewide 80,000 Camera Scalability Blueprint

Enterprise hardware, network sizing, and high availability

- Hardware BOM: 66 2U Edge Servers (2 per District) + 264 NVIDIA L4 GPUs statewide.
- Tiered Storage: Hot (7-day NVMe edge) to Warm (90-day object store) to Cold (7-year archive).
- Active-Active DC-DR: Dual datacenter (Gandhinagar + Vadodara) with RPO < 1s, RTO < 15s.
- District Autonomy: District edge servers continue local ANPR matching during WAN outage.

SentinelGrid — End-to-End Workflow & Integration Dataflow
From Optical CCTV Stream Ingestion to Live Tactical PCR Interception & Court Evidence

